# IoT Base NBIoT-CN

SKU:K064




















## Description

**IoT Base NBIoT-CN** is a functional base specifically designed for IoT NB-IoT data transmission scenarios. The kit comes with a powerful core controller BASIC (built-in ESP32, integrated Wi-Fi), and integrates rich interface resources (I2C, UART, GPIO, RS485, etc.) to adapt to various industrial field protocols (such as ModBus). It has a built-in SIM7020C module, mainly covering the Chinese Cat-NB frequency band, and integrates an SMA external antenna interface to improve communication quality and signal stability. It comes with a multi-functional rail fixing base, perfectly embedded in industrial application scenarios.

## Features

- SIM7020C/Suitable for China
- Strong signal access capability
- AT command control
- SIM card type: MicroSIM
- External antenna: SMA antenna
- Serial communication: UART 115200bps
- RS485 communication interface (with 12V input, built-in DC-DC 12V to 5v)
- Multi-functional rail fixing base included
- NB module status indicator
- BASIC host included
- Grove expansion interface:
  - I2C x1
  - GPIO x1
  - UART x1
- SIM card slot specification: MicroSIM
- Cat-NB frequency band:
  - B1/B3/B5/B8
- Data transmission (kbps):
  - 26.15(DL)/62.5(UL)
- Network protocols:
  - TCP/UDP/LWM2M/COAP/ MQTT/HTTP/HTTPS/ TLS/DTLS/DNS/NTP/ PING/OneNET/Telecom Cloud/Ayla Cloud*

## Includes

- 1 x IoT Base NBIoT-CN
- 1 x BASIC controller
- 1 x 12V DC power adapter
- 1 x VH-3.96-2P terminal
- 1 x VH-3.96-4P terminal
- 1 x Din-mount rail fixing base
- 1 x SMA antenna
- 2 x M3*28 screws
- 1 x M3 hex wrench
- 1 x NB-IoT data card (Mobile NB data: 300M)

## Applications

- Smart metering
- Remote monitoring
- Shared bikes

## Specifications

| Specification | Parameters |
| --- | --- |
| Communication module | SIM7020C |
| Supported Cat-NB bands | B1/B3/B5/B8 |
| Data transfer speed | 26.15(DL)/62.5(UL) |
| Network protocols | TCP/UDP/LWM2M/COAP/ MQTT/HTTP/HTTPS/ TLS/DTLS/DNS/NTP/ PING/OneNET/Telecom Cloud/Ayla Cloud* |
| Interface resources | I2C, UART, GPIO, RS485 (supports ModBus Master/Slave) |
| Power supply | USB power supply / 12V DC power adapter |
| Communication method | UART 115200bps |
| Net weight | 71.3g |
| Gross Weight | 197g |
| Product Size | 80 x 54 x 33mm |
| Package Size | 98 x 106 x 57mm |

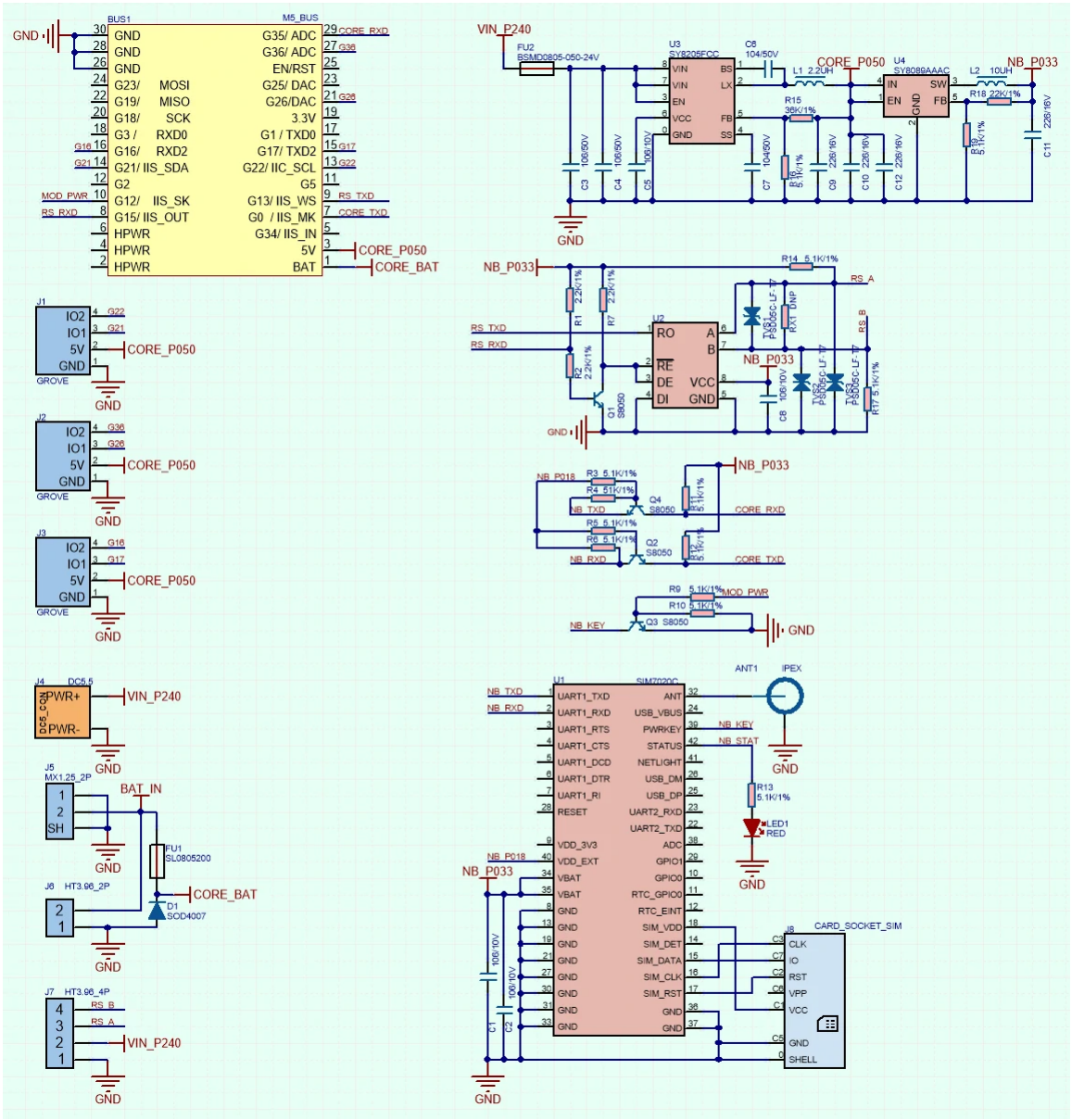
# Certifications

- RoHS/REACH/CCC/CTA/SRRC/Mobile/Unicom/Telecom/ATEX

## Carrier Certification

- China Telecom/China Unicom/China Mobile

# Schematics




# PinMap

## RS485

| CORE | G15 | G13 | 5V | GND |
|---|---|---|---|---|
| RS485 | TX | RX | VIN | GND |

## SIM7020C

| CORE | G0 | G35 | G12 |
|---|---|---|---|
| SIM7020C | NB-RXD | NB-TXD | PWRKEY |

## HY2.0-4P

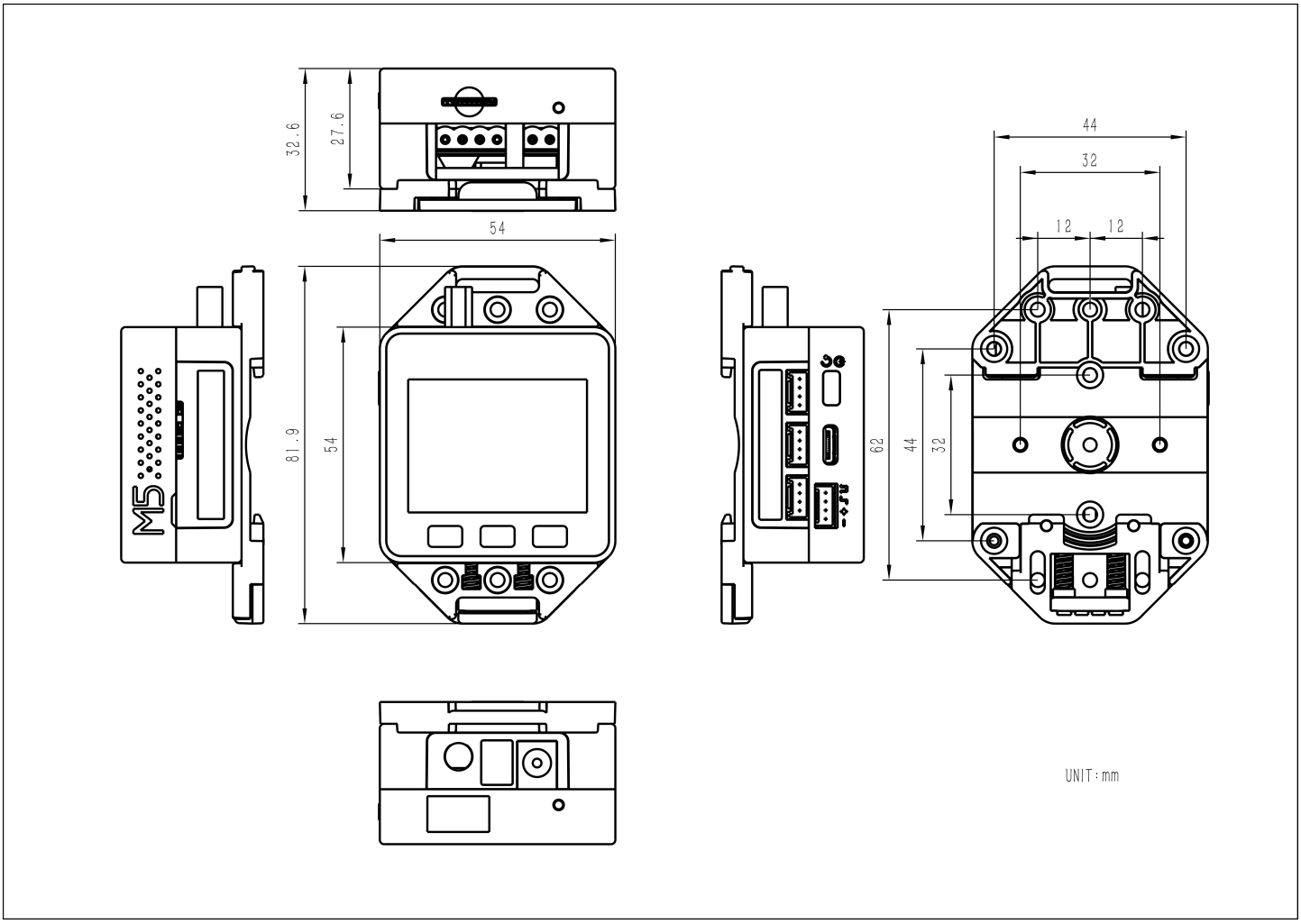
| CORE | G21 | G22 | 5V | GND |
|---|---|---|---|---|
| I2C PORT | SDA | SCL | VIN | GND |

| CORE | G17 | G16 | 5V | GND |
| --- | --- | --- | --- | --- |
| UART PORT | TX | RX | VIN | GND |

| CORE | G36 | G26 | 5V | GND |
| --- | --- | --- | --- | --- |
| GPIO PORT | ADC | DAC | VIN | GND |

## Model Size

- IoT Base NBIoT-CN Model Size PDF




## Datasheets

- SIM7020 Series datasheet
- SIM7020 Series_Ayla_Application Note_V1.03
- SIM7020 Series_CTBURST_Application Note_V1.01
- SIM7020 Series_CoAP_Application Note_V1.03
- SIM7020 Series_EAT_Environment & Compilation & Burning Guide_V1.02.
- SIM7020 Series_FOTA_Application_Note_V1.02
- SIM7020 Series_HTTP(S)_Application Note_V1.04
- SIM7020 Series_LWM2M_Application Note_V1.03
- SIM7020 Series_Low Power Mode_Application Note_V1.05
- SIM7020 Series_MQTT(S)_Application Note_V1.05
- SIM7020 Series_NVRAM_Application Note_V1.02

- SIM7020 Series_SAT_Application Note_V1.01
- SIM7020 Series_SNTP_Application Note_V1.03
- SIM7020 Series_TCPIP_Application Note_V1.04

# Softwares

## Arduino

- IoT Base NBIoT MQTT Client Example
- IoT Base NBIoT ModBus RTU Master Example
- IoT Base NBIoT ModBus RTU Slave Example
- IoT Base NBIoT_Factory Test
- ArduinoModbus Library
- Arduino485 Library
- TinyGSM Library

## UiFlow1

- IoT Base NBIoT UiFlow1 Docs

## Protocol

- SIM7020 Series_AT Command Manual_V1.05

# Video

IoT_BASE_NB_CN.mp4